

EFY Note

The source code of this project is available for free download at source.efymag.com

```
#define WLAN_PASS "xxxxxxx" //password
                                here
#define AIO_SERVER "io.adafruit.com"
#define AIO_SERVERPORT 1883 // use 8883 is
                                used for SSL
#define AIO_USERNAME "xxxxxxxxxx" //User
                                name of Adafruit IO (for this click the
                                Golden Key in Adafruit IO Dashboard)
#define AIO_KEY "xxxxxxxxxx" //key of
                                Adafruit IO (for this click the Golden
                                Key in Adafruit IO Dashboard)
```

Enter your feed name to subscribe to MQTT server. For more details on MQTT, visit <http://mqtt.org/>

Subscribe to MQTT. It is done in the code in Void Setup() as shown below.

```
void setup()
{
  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, LOW);
  Serial.begin(9600);
  delay(10);
  Serial.println();
  Serial.println();
  Serial.print("Connecting to "); //
  Connect to WiFi access point.
  Serial.println(WLAN_SSID);
  WiFi.begin(WLAN_SSID, WLAN_PASS);
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  Serial.println();
  Serial.println("WiFi connected");
  Serial.println("IP address: ");
  Serial.println(WiFi.localIP());
  mqtt.subscribe(&onoffbutton); // Setup
  MQTT subscription for onoff feed.
}
```

relayPin is set as output, which is initially low. This output is defined in Void Loop(), as shown below.

```
void loop()
{
  // Ensure the connection to the MQTT
  server is alive (this will make the first
  // connection and automatically reconnect
  when disconnected). See the MQTT_connect
  // function definition further below.
```

DO-IT-YOURSELF

```
MQTT_connect();
// this is our 'wait for incoming
subscription packets' busy subloop
// try to spend your time here
Adafruit_MQTT_Subscribe *subscription;
while ((subscription = mqtt.readSubscription(5000)))
{
  if (subscription == &onoffbutton) {
    Serial.print(F("Got: "));
    Serial.println((char *)onoffbutton.
lastread);
    String response = (char*)onoffbutton.
lastread; //converts the received 1
or 0 to string to compare in the if-
else statement
    if (response == "1")
    {
      digitalWrite(relayPin, HIGH);
    }
    else
    {
      digitalWrite(relayPin, LOW);
    }
  }
}
```

Save the code and upload it via USB cable. Do not forget to select the COM port from Tools menu in Arduino IDE before uploading the code.

Testing chatbot

After connecting the circuit diagram and the electrical appliance, such as AC bulb, power on the circuit.



Fig. 25: Demonstration of chatbot prototype

Open Facebook Messenger on your phone. Search for the chatbot with its name and start chatting with your IoT system by giving voice commands like 'Turn on the light' or 'Switch on the light.' Demonstration of the chatbot prototype is shown in Fig. 25. **EFY**

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-Morris Hite

